

Centre Number	Candidate Number	Name
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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Advanced Level

PHYSICS

9702/04

Paper 4

May/June 2005

1 hour

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen in the spaces provided on the Question Paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
The number of marks is given in brackets [] at the end of each question or part question.
You may lose marks if you do not show your working or if you do not use appropriate units.

For Examiner's Use	
1	
2	
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Total	

If you have been given a label, look at the details. If any details are incorrect or missing, please fill in your correct details in the space given at the top of this page.

Stick your personal label here, if provided.

This document consists of **16** printed pages.



Data

speed of light in free space,

$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$

permeability of free space,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$$

permittivity of free space,

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$$

elementary charge,

$$e = 1.60 \times 10^{-19} \text{ C}$$

the Planck constant,

$$h = 6.63 \times 10^{-34} \text{ J s}$$

unified atomic mass constant,

$$u = 1.66 \times 10^{-27} \text{ kg}$$

rest mass of electron,

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

rest mass of proton,

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

molar gas constant,

$$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

the Avogadro constant,

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

the Boltzmann constant,

$$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

gravitational constant,

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

acceleration of free fall,

$$g = 9.81 \text{ m s}^{-2}$$

Formulae

uniformly accelerated motion,

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas,

$$W = p\Delta V$$

gravitational potential,

$$\phi = -\frac{Gm}{r}$$

simple harmonic motion,

$$a = -\omega^2 x$$

velocity of particle in s.h.m.,

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

resistors in series,

$$R = R_1 + R_2 + \dots$$

resistors in parallel,

$$1/R = 1/R_1 + 1/R_2 + \dots$$

electric potential,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series,

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel,

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor,

$$W = \frac{1}{2} QV$$

alternating current/voltage,

$$x = x_0 \sin \omega t$$

hydrostatic pressure,

$$p = \rho gh$$

pressure of an ideal gas,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

radioactive decay,

$$x = x_0 \exp(-\lambda t)$$

decay constant,

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

critical density of matter in the Universe,

$$\rho_0 = \frac{3H_0^2}{8\pi G}$$

equation of continuity,

$$Av = \text{constant}$$

Bernoulli equation (simplified),

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Stokes' law,

$$F = Ar\eta v$$

Reynolds' number,

$$R_e = \frac{\rho v r}{\eta}$$

drag force in turbulent flow,

$$F = Br^2 \rho v^2$$

Answer **all** the questions in the spaces provided.

- 1 The orbit of the Earth, mass 6.0×10^{24} kg, may be assumed to be a circle of radius 1.5×10^{11} m with the Sun at its centre, as illustrated in Fig. 1.1.

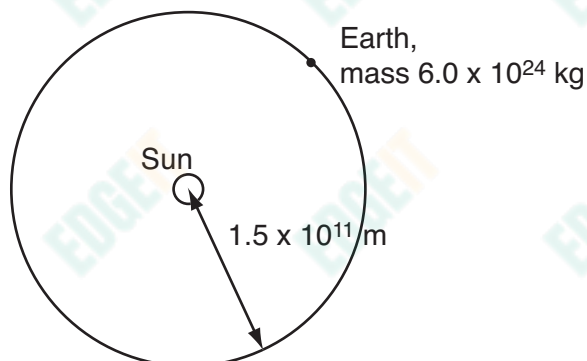


Fig. 1.1

The time taken for one orbit is 3.2×10^7 s.

(a) Calculate

- (i) the magnitude of the angular velocity of the Earth about the Sun,

angular velocity = rad s^{-1} [2]

- (ii) the magnitude of the centripetal force acting on the Earth.

force = N [2]

- (b) (i) State the origin of the centripetal force calculated in (a)(ii).

.....
..... [1]

- (ii) Determine the mass of the Sun.

mass = kg [3]

- 2 (a) State what is meant by an *ideal* gas.

.....

[2]

- (b) The product of pressure p and volume V of an ideal gas of density ρ at temperature T is given by the expressions

$$p = \frac{1}{3}\rho\langle c^2 \rangle$$

and $pV = NkT,$

where N is the number of molecules and k is the Boltzmann constant.

- (i) State the meaning of the symbol $\langle c^2 \rangle$.

.....[1]

- (ii) Deduce that the mean kinetic energy E_K of the molecules of an ideal gas is given by the expression

$$E_K = \frac{3}{2}kT.$$

[2]

- (c) In order for an atom to escape completely from the Earth's gravitational field, it must have a speed of approximately $1.1 \times 10^4 \text{ m s}^{-1}$ at the top of the Earth's atmosphere.

- (i) Estimate the temperature at the top of the atmosphere such that helium, assumed to be an ideal gas, could escape from the Earth. The mass of a helium atom is $6.6 \times 10^{-27} \text{ kg}$.

temperature = K [2]

- (ii) Suggest why some helium atoms will escape at temperatures below that calculated in (i).

.....
[1]

- 3 (a) Define *specific latent heat of fusion*.

.....

.....

.....[2]

- (b) A mass of 24 g of ice at -15°C is taken from a freezer and placed in a beaker containing 200 g of water at 28°C . Data for ice and for water are given in Fig. 3.1.

	specific heat capacity / $\text{J kg}^{-1} \text{K}^{-1}$	specific latent heat of fusion / J kg^{-1}
ice	2.1×10^3	3.3×10^5
water	4.2×10^3	—

Fig. 3.1

- (i) Calculate the quantity of thermal energy required to convert the ice at -15°C to water at 0°C .

energy = J [3]

- (ii) Assuming that the beaker has negligible mass, calculate the final temperature of the water in the beaker.

temperature = $^{\circ}\text{C}$ [3]

- 4 A tube, closed at one end, has a constant area of cross-section A . Some lead shot is placed in the tube so that the tube floats vertically in a liquid of density ρ , as shown in Fig. 4.1.

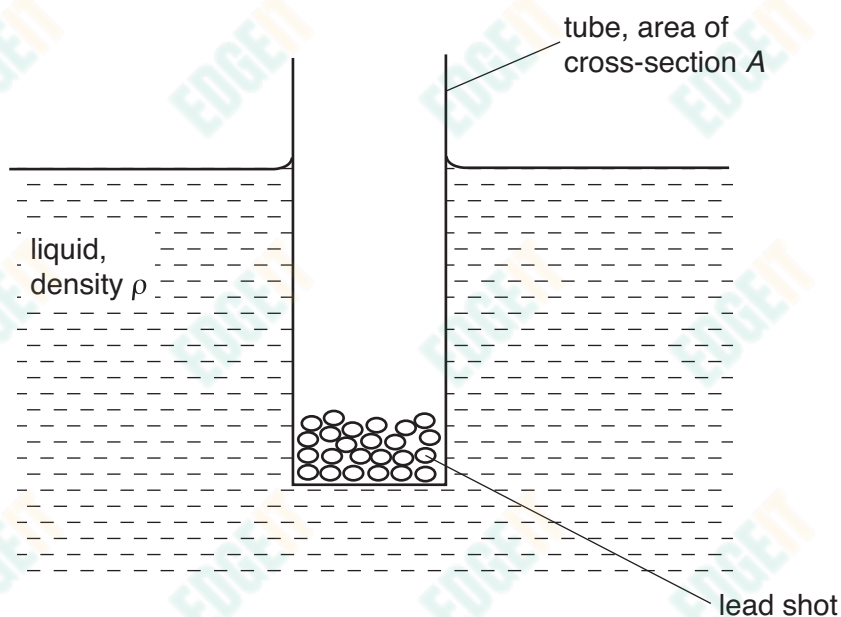


Fig. 4.1

The total mass of the tube and its contents is M .

When the tube is given a small vertical displacement and then released, the vertical acceleration a of the tube is related to its vertical displacement y by the expression

$$a = -\frac{A\rho g}{M} y,$$

where g is the acceleration of free fall.

- (a) Define *simple harmonic motion*.

.....

.....

.....[2]

- (b) Show that the tube is performing simple harmonic motion with a frequency f given by

$$f = \frac{1}{2\pi} \sqrt{\frac{A\rho g}{M}}.$$

[3]

- (c) Fig. 4.2 shows the variation with time t of the vertical displacement y of the tube in another liquid.

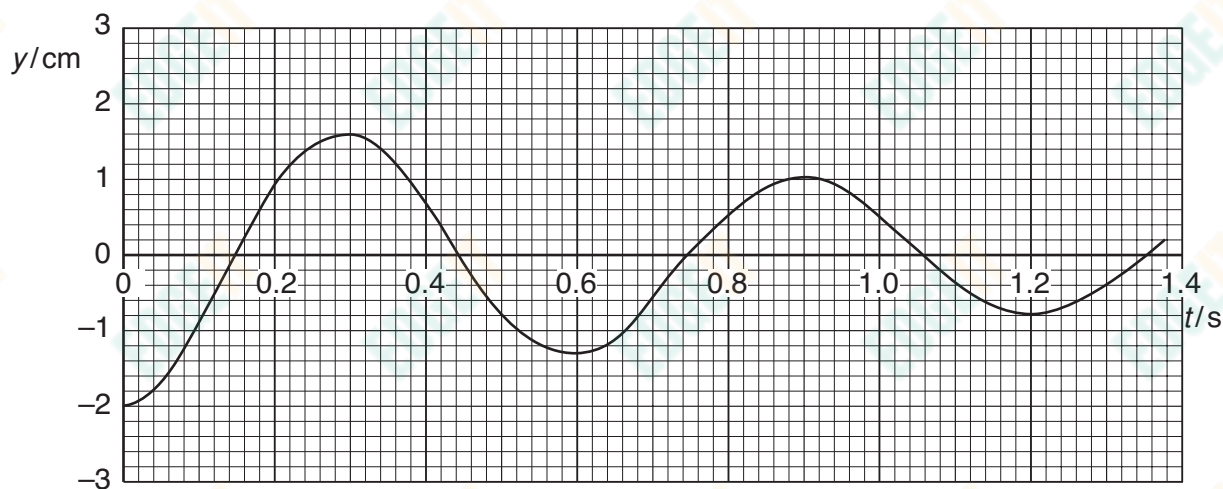


Fig. 4.2

- (i) The tube has an external diameter of 2.4 cm and is floating in a liquid of density 950 kg m^{-3} . Assuming the equation in (b), calculate the mass of the tube and its contents.

mass = kg [3]

- (ii) State what feature of Fig. 4.2 indicates that the oscillations are damped.

.....
 [1]

- 5 An isolated conducting sphere of radius r is given a charge $+Q$. This charge may be assumed to act as a point charge situated at the centre of the sphere, as shown in Fig. 5.1.

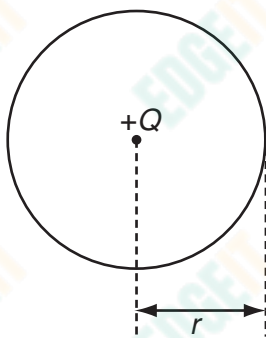


Fig. 5.1

Fig. 5.2. shows the variation with distance x from the centre of the sphere of the potential V due to the charge $+Q$.

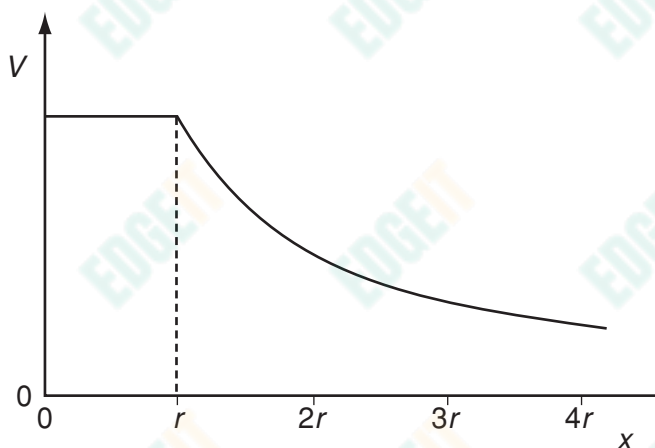


Fig. 5.2

- (a) State the relation between electric field and potential.

.....[1]

- (b) Using the relation in (a), on Fig. 5.3 sketch a graph to show the variation with distance x of the electric field E due to the charge $+Q$.

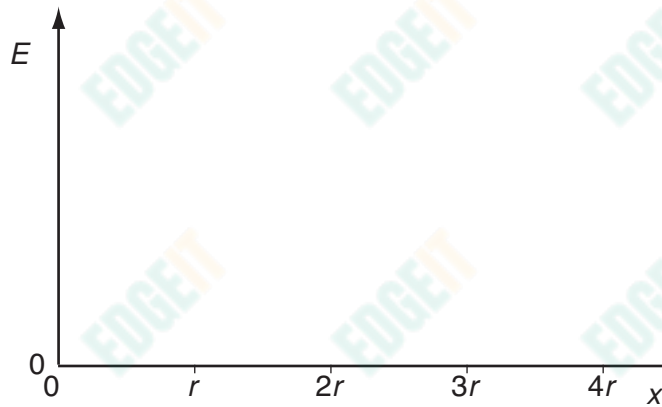


Fig. 5.3

[3]

- 6 An ideal iron-cored transformer is illustrated in Fig. 6.1.

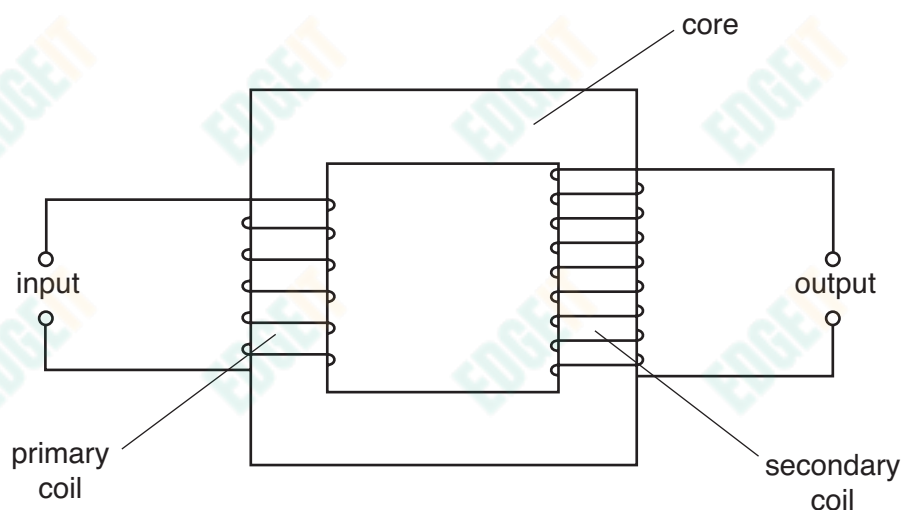


Fig. 6.1

(a) Explain why

- (i)** the supply to the primary coil must be alternating current, not direct current,

.....
.....
.....[2]

- (ii)** for constant input power, the output current must decrease if the output voltage increases.

.....
.....
.....[2]

- (b) Fig. 6.2 shows the variation with time t of the current I_p in the primary coil. There is no current in the secondary coil.

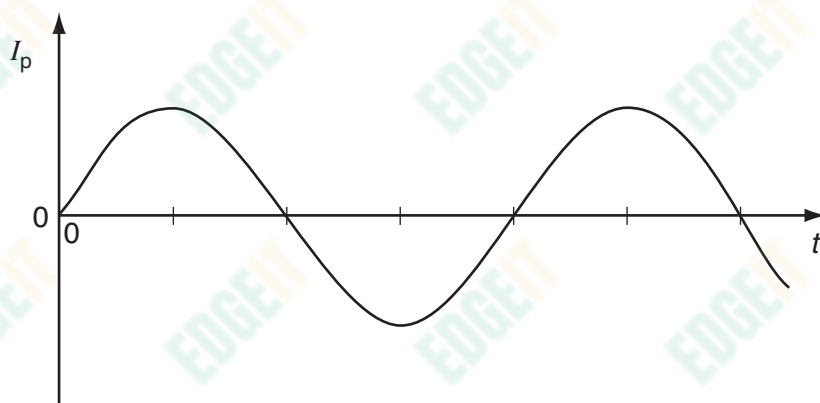


Fig. 6.2



Fig. 6.3



Fig. 6.4

- (i) Complete Fig. 6.3 to show the variation with time t of the magnetic flux Φ in the core. [1]
- (ii) Complete Fig. 6.4 to show the variation with time t of the e.m.f. E induced in the secondary coil. [2]
- (iii) Hence state the phase difference between the current I_p in the primary coil and the e.m.f. E induced in the secondary coil.

phase difference = [1]

- 7 The isotope Manganese-56 decays and undergoes β -particle emission to form the stable isotope Iron-56. The half-life for this decay is 2.6 hours. Initially, at time $t = 0$, a sample of Manganese-56 has a mass of $1.4\ \mu\text{g}$ and there is no Iron-56.

- (a) Complete Fig. 7.1 to show the variation with time t of the mass of Iron-56 in the sample for time $t = 0$ to time $t = 11$ hours.

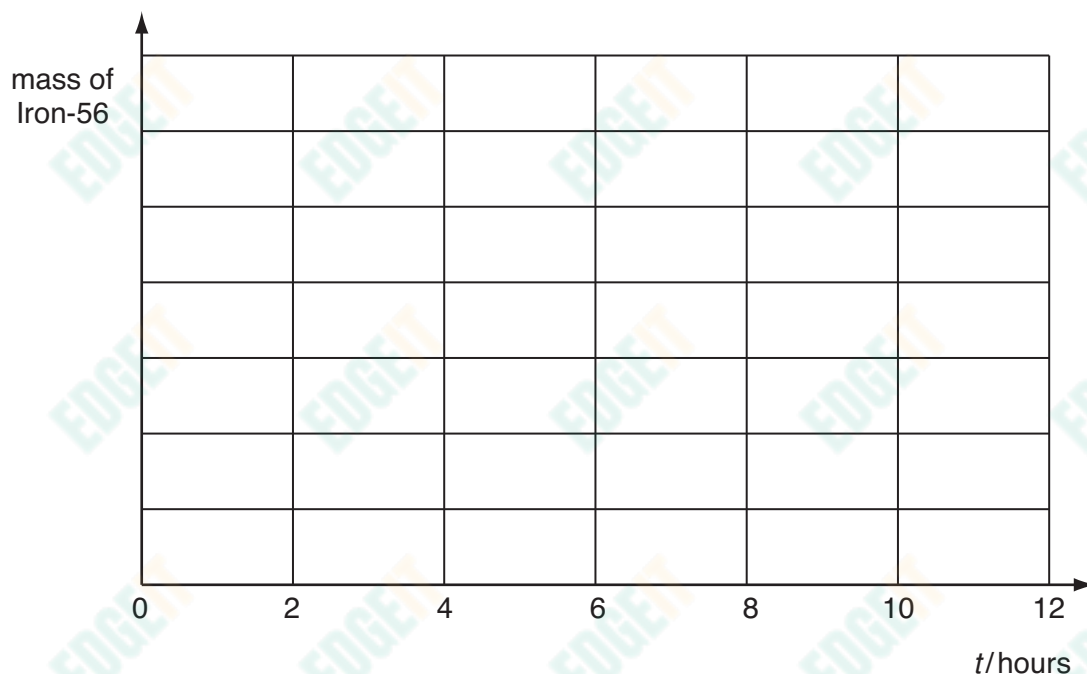


Fig. 7.1

[2]

- (b) For the sample of Manganese-56, determine
- (i) the initial number of Manganese-56 atoms in the sample,

number =[2]

- (ii) the initial activity.

activity = Bq [3]

(c) Determine the time at which the ratio

$$\frac{\text{mass of Iron-56}}{\text{mass of Manganese-56}}$$

is equal to 9.0.

time = hours [2]

8 (a) Define *capacitance*.

.....
.....[1]

(b) (i) One use of a capacitor is for the storage of electrical energy.
Briefly explain how a capacitor stores energy.

.....
.....
.....[2]

(ii) Calculate the change in the energy stored in a capacitor of capacitance $1200\ \mu\text{F}$
when the potential difference across the capacitor changes from 50 V to 15 V.

energy change = J [3]